

ALEXANDER LU

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SUMMARY

EECS student at UC Berkeley (4.00 GPA) with SWE internships at Apple and Amazon, an ACL 2026 publication, and ML research across BAIR and UCSF. Built and shipped systems at scale: 300K+ TPM traffic infrastructure, 95%-accuracy seizure detection, and RL pipelines fine-tuning 9 reasoning models.

EDUCATION

University of California, Berkeley

B.S. Electrical Engineering and Computer Science, Mathematics Minor

Aug. 2024 – May 2028

GPA: 4.00/4.00

- Courses: Data Structures, Algorithms, Computer Architecture, Linear Algebra, Discrete Math, Stochastic Processes, Signal Processing, Real Analysis, Circuit Analysis

SKILLS

Languages: C/C++, Java, Python, Rust, TypeScript/HTML/CSS, SQL, Swift

Frameworks & Tools: NumPy, Pandas, PyTorch, TensorFlow, Scikit-Learn, VERL, vLLM, TransformerLens, SciPy, FAISS, React.js, Flask, Spring Boot, Node.js, Docker, AWS, Git, Linux

Concepts: Reinforcement Learning, Distributed Training, LLM Fine-Tuning, Mechanistic Interpretability, Time Series Analysis, Statistical Modeling, Signal Processing, Distributed Systems, Numerical Optimization, Algorithms, Data Structures, Computer Architecture

EXPERIENCE

Apple – Hardware Automation

Apr. 2026 – Present

Applied AI SWE Intern

- Building heatmap visualizer to statistically determine regressions and improvements across **300+** software and hardware builds
- Implementing unsupervised time-series anomaly detection and root cause analysis across hardware data from **200+** devices
- Training robotic manipulator with RL in Issac Sim and bridging Sim2Real gap via domain randomization.

Amazon – Traffic Engineering

Jan. 2026 – Apr. 2026

Software Development Engineering Intern

- Built an agentic AI orchestration on AWS to automate CDN deployments across **20+** global regions.
- Enabled safe, human-in-the-loop traffic shifts at **300K+ TPM** in under **10 minutes** with GenAI monitoring and validation.
- Reduced deployment time by **59%** and cut potential human error points by **82%**, improving production reliability.

Berkeley AI Research – Berkeley Speech Group

May 2025 – Present

Machine Learning Researcher

- Authored paper accepted to **ACL 2026** showing Evolutionary Strategies induce catastrophic forgetting vs. GRPO via denser weight updates and higher ℓ_2 norms.
- Fine-tuned **9** Qwen3 reasoning models with GRPO and ES, improving faithfulness metrics from **80% to 88%**; contributed logging and training features to the open-source **VERL** RL library.
- Building AlphaEvolve-style continual learning harnesses to surpass SOTA on open mathematical problems via inference-time training.
- Implemented PUCT-based tree search sampler enabling LLMs to continuously evolve programs across a structured search tree.

UCSF Weill Institute for Neurosciences – Chen Lab

Jan. 2025 – Dec. 2025

Machine Learning Intern

- Employed signal processing methods (Wavelet Transform, Fourier Analysis) with SciPy for EEG feature extraction.
- Engineered seizure detection pipeline with EEG time series using SVM and GMM models; achieved **95%** accuracy.
- Trained a Keypoint MoSeq pipeline to perform unsupervised learning on detecting rodent movements.
- Optimized memory usage in a seizure-detection Python library from **200GB+** down to **4GB** via chunking.

PROJECTS

EigenAgent | github.com/YLu-1258/eigenAgent

- Built a fully local LLM desktop assistant using Rust and llama.cpp for offline, on-device inference with GGUF-quantized models.
- Implemented token-level streaming via an event-driven pipeline, improving responsiveness and reducing latency.
- Integrated MCP-based tool invocation to support agentic workflows and extensible custom tool servers.

Semantic Backlinks Generator | [GitHub](https://github.com)

- Python backlink generator for Obsidian-style blogs using a sentence transformer for semantic encoding.
- Employed FAISS for efficient similarity search to surface semantically related posts and reduce manual linking effort.
- Built a data-visualization pipeline for customizing density, confidence level, and semantic weights on a graph.

Mechanistic Interpretability Toolkit | [GitHub](https://github.com)

- Engineered a mechanistic interpretability toolkit for GPT-2 and Qwen3 supporting token attribution, attention head analysis, and activation patching via TransformerLens, FastAPI, and React.js.
- Implemented activation patching across 28 layers and 24 heads to identify causal circuits with real-time Δ Logit heatmap visualization.
- Streamed layer-wise logit lens predictions via WebSocket, visualizing the transformer layers where next-token predictions stabilize.

PUBLICATIONS

Dec. 2025 – Evolutionary Strategies lead to Catastrophic Forgetting in LLMs – Accepted into ACL 2026 main conference